# Credit Card Fraud Detection Using Deep Learning

An MLP + Autoencoder Stacking Ensemble Approach

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**ETA3_AI — AI Training Program (Huawei - NTI)**

**Instructor:** Dr. Ahmed Arafa

**Team Leader:** Shorouk Yehia Ismail

**Team Members:**

- Amr Mohamed.
- Saeed Seliem.
- Jana Hazem.
- Omnia Hesham.

# Fraud is rare enough to make accuracy misleading

DATASET

283,726

transactions after duplicate removal

473

fraud · 0.167%

**Accuracy alone is misleading.** Predicting legitimate for almost every transaction can look excellent while missing the fraud cases that matter.

EVALUATION LENSES

Precision · Recall · F1

ROC-AUC · PR-AUC · False positives

# Thirty engineered features make the input reproducible

The same feature order is used in training and deployment.

1

## V1–V28

28 anonymized PCA features preserve the transformed transaction signal.

2

## Amount_log

Amount_log = log1p(Amount), reducing extreme-value skew.

3

## Hour

**Hour = (Time // 3600) % 24**
Raw Time and Amount are removed; Amount is transformed into Amount_log using log1p.

# Data split and leakage control

## Stratified split

Train: 181,584
Validation: 45,396
Test: 56,746

## Scale

StandardScaler is fitted only on training data, then applied to validation and test.

## Balance

SMOTE is applied only to training data. Validation and test retain the real class distribution.

# Overall system architecture

Two complementary signals are calibrated and combined into one final decision.

INPUT

**30**

engineered features

**MLP**

Fraud probability

**Autoencoder**

Reconstruction error

META-MODEL

Meta-scaler → Logistic Regression → threshold 0.7446

30 features → MLP + Autoencoder → meta-features → ensemble score → Fraud / Legitimate

# The MLP learns a non-linear fraud boundary

ARCHITECTURE

**30 → 128 → 64 → 32 → 16 → 1**

ReLU hidden layers · sigmoid output

Dropout: 0.40 → 0.30 → 0.20

## Training controls

SMOTE-balanced training

EarlyStopping on validation PR-AUC

ReduceLROnPlateau

Threshold selected by maximizing **Recall** on validation — not by maximizing F1.

# The MLP catches the most fraud, at a high false-alarm cost

MLP TEST RESULTS · THRESHOLD 0.01

| | |
|---|---|
| Accuracy | 99.85% |
| Precision | 53.42% |
| Recall | 82.11% |
| F1 | 64.73% |
| ROC-AUC | 95.22% |
| PR-AUC | 79.86% |
| False positives | 68 |

WHY IT MATTERS

82.11%

highest Recall

The MLP also has the highest PR-AUC, but its low threshold creates more false alarms.

# The Autoencoder is precise, with fewer false alarms

AUTOENCODER TEST RESULTS · THRESHOLD 6.6521

Threshold selected on the validation set by maximizing F2-score

| | |
|---|---|
| Accuracy | 99.92% |
| Precision | 80.23% |
| Recall | 72.63% |
| F1 | 76.24% |
| ROC-AUC | 94.05% |
| PR-AUC | 65.01% |
| False positives | 17 |

READOUT

69 / 95

true positives

Precision is stronger than the MLP, with fewer false alarms.

# The Autoencoder learns normality, then flags deviation

It learns the structure of normal transactions and scores how far an input deviates.

ARCHITECTURE

**30 → 32 → 16 → 8 → 16 → 32 → 30**

Normal transactions only · RobustScaler

Gaussian noise · L2 regularization

SIGNAL

Weighted reconstruction error flags anomalous behavior.

Fraud-correlated features include V17, V14, V12, V10.

# Stacking turns two signals into one decision

The meta-model sees both the MLP probability and the Autoencoder reconstruction error.

META-FEATURES

**MLP fraud probability**

**Autoencoder reconstruction error**

CALIBRATION

Meta-scaler: StandardScaler

Meta-model: Logistic Regression

class_weight="balanced"

FINAL DECISION

0.7446

ensemble threshold

Fraud / Legitimate

# The ensemble improves the operating balance

ENSEMBLE TEST RESULTS · THRESHOLD 0.7446

| | |
|---|---|
| Accuracy | 99.94% |
| Precision | 86.90% |
| Recall | 76.84% |
| F1 | 81.56% |
| ROC-AUC | 95.72% |
| False positives | 11 |

READOUT

73 / 95

true positives

Best overall balance and fewest false positives on the held-out test set.

# Model comparison

The ensemble is the balance choice, not a universal winner.

| Model | Precision | Recall | F1 | ROC-AUC | PR-AUC | FP |
|---|---|---|---|---|---|---|
| MLP | 53.42% | **82.11%** | 64.73% | 95.22% | **79.86%** | 68 |
| Autoencoder | 80.23% | 72.63% | 76.24% | 94.05% | 65.01% | 17 |
| Ensemble | **86.90%** | 76.84% | **81.56%** | **95.72%** | 72.42% | **11** |

> **Ensemble: best overall balance and fewest false positives. MLP: highest Recall and PR-AUC.**

# Streamlit deployment

Inference mirrors the training-time pipeline without shipping the training dataset.

INFERENCE FLOW

User inputs transaction features

→ same preprocessing

→ MLP probability

→ Autoencoder reconstruction error

→ ensemble meta-model

→ final threshold

→ **FRAUD / LEGITIMATE**

DEPLOYMENT RULES

Saved artifacts only.

Does not load the training dataset.

Does not retrain models.

FINAL TAKEAWAY

# Deep learning can detect highly imbalanced fraud patterns.

MLP provides strong fraud sensitivity.

Autoencoder detects deviations from normal behavior.

Stacking combines both signals.

Ensemble achieved the best overall balance on the held-out test set.

Final deployment is available through Streamlit.